

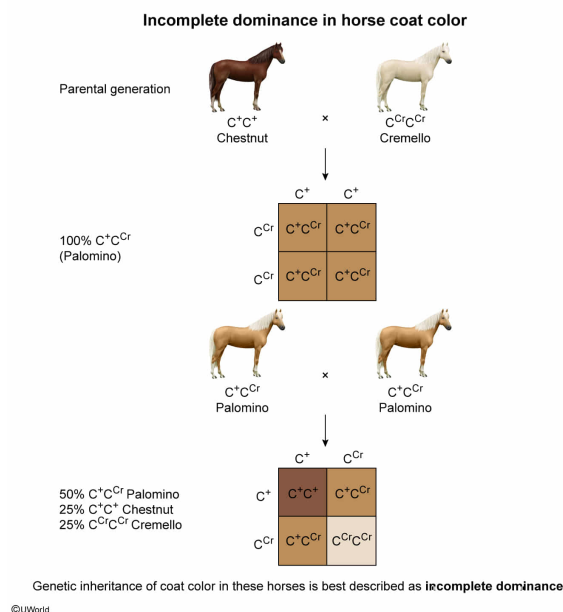
Palomino (golden brown) horses are bred for their coat coloration. A breeder will achieve 100% Palomino offspring by crossing a chestnut (dark brown) horse with a cremello (cream) horse. However, if a breeder crosses two Palomino horses, only 50% of the offspring will be Palomino, while 25% will be chestnut and 25% will be cremello. Genetic inheritance of coat color in these horses is best described as:

- ☐ A. codominance. [21%]
- ☐ B. complete dominance. [8%]
- ☐ C. variable expressivity. [12%]
- ✓ ☒ D. incomplete dominance. [57%]

Correct

58% answered correctly

Explanation:



For traits inherited in a **completely dominant** fashion, the **dominant** phenotype **masks** the **recessive** phenotype in a **heterozygous** individual. However, in some cases, both **alleles** are expressed phenotypically, resulting in heterozygote with combined or apparently blended traits. This situation is referred to as **incomplete dominance**.

In this question, Palomino (golden brown) horses are bred for their coat color. If a breeder crosses a chestnut (dark brown) horse with a cremello (cream) horse, 100% of the offspring will be Palomino. This suggests that chestnut and cremello horses are both homozygous for coat color, and that Palomino horses are heterozygous.

Furthermore, the question states that if two Palomino horses are crossed, 50% of the offspring will be Palomino (heterozygous), while 25% will be chestnut and 25% will be cremello. Because crossing Palominos produces chestnut and cremello offspring, the heterozygous phenotype is not due to simple blending of the dominant and recessive

traits.

Further, because the Palomino coloration is *distinct* from either the chestnut or cremello phenotype, this is *not* a case of complete dominance, in which the heterozygote would be either chestnut or cremello, depending on which allele was dominant **(Choice B)**. Rather, this is an example of **incomplete dominance**, in which *neither* the chestnut allele nor the cremello allele is dominant. In this case, they are **incompletely dominant** to one another, producing heterozygotes (Palomino horses) with an **intermediate phenotype**.

(Choice A) In **codominance**, neither allele for a trait is dominant to the other. However, unlike in incomplete dominance, *both* phenotypes appear in heterozygotes. Because Palomino horses display a phenotype *unlike* either parental phenotype, rather than *both* parental phenotypes, these horses are an example of incomplete dominance, *not* codominance.

(Choice C) **Variable expressivity** refers to a single genotype being able to produce distinct organisms with distinct phenotypes. In contrast, constant expressivity refers to a single genotype producing organisms that express the same phenotype. In the situation presented in the question, each genotype produces a *distinct* phenotype: homozygous chestnut produces chestnut coloration; homozygous cremello produces cremello coloration; and the heterozygote produces Palomino coloration. Therefore, this is *not* a case of variable expressivity.

Educational objective:

In incomplete dominance, neither of two alleles is completely dominant over the other, producing heterozygotes with intermediate or blended phenotypes of the homozygous parental generation.

Time spent:0 sec

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